

Homework 9

due 13.12.2023, 14:00

Exercise 9.1 - Advanced MPI (10 points)

In this exercise we practice parallel I/O and latency hiding.

The data to be read or written in this exercise are $M \times N$ matrices A stored in the following binary format:

M as a 4-byte little-endian unsigned integer

N as a 4-byte little-endian unsigned integer

a_{00} as an 8 byte little-endian double

a_{01} as an 8 byte little-endian double

$\vdots$

$a_{0,M-1}$ as an 8 byte little-endian double

$\vdots$

$a_{M-1,N-1}$ as an 8 byte little-endian double

The file size in total is $4 + 4 + 8MN$ bytes. Every process keeps a $m \times N$ portion of A in its local memory, with $m = M/\text{nproc}$

- Use a block-row parallelization and write functions to read/write such matrices using parallel-I/O.
- Write a function that “smears” a matrix, i.e. a new matrix B is computed, with components

$$b_{ij} = \frac{1}{1 + 4\kappa} (a_{ij} + \kappa[a_{i+1,j} + a_{i-1,j} + a_{i,j+1} + a_{i,j-1}]) ,$$

where periodic boundaries are used, i.e. $a_{m,j} \equiv a_{0,j}$ etc. With the block-row parallelization this requires the exchange of rows between neighboring processes. Use latency hiding for this exchange, i.e. start non-blocking sending and receiving, and compute the rows $i \in 1, \dots, m-2$ of the local B matrix while the communication is going on. When the communication is finished, compute rows 0 and $m-1$.

- The smearing can be iterated, i.e. the result can be copied over the original A matrix and smeared again.
- As an application of your code, load the matrix from `/work/korzec/LAB2/ex9/matrix.bin` on stromboli. Apply 5, 20 or 100 iterations of smearing with $\kappa = 0.1$ and write out a binary file for each case. Visualize your binary files using the matlab script `matview.m` (just adjust the file names) and include the resulting images in your PDF.